

Utility and Money

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Money Is Not Utility

- Money is the most familiar unit for measuring value in decisions.
- But dollars and utility are **not the same thing**.
- Today we explore how they differ, and why this matters.

Associated Reading

Odds and Ends, Sections 12.5 and 13.1

Getting $\$2x$ is not twice as valuable as getting $\$x$.

Why? Because getting $\$2x$ is like getting $\$x$, *then getting another $\$x$ once you're already richer*. And the second $\$x$ is worth less to you than the first.

This is **declining marginal utility**: the extra value of each additional dollar decreases as you get richer.

These two claims are both true, and closely connected:

1. You're better off getting a million dollars than getting a 50/50 shot at two million dollars.
2. Getting a million dollars changes your life more than it changes a billionaire's life.

Both are grounded in declining marginal utility.

The graph of utility as a function of wealth should satisfy:

- **More money $\rightarrow$ more utility** (the curve slopes upward).
- **Declining marginal utility** (the curve bends downward; it's concave).

Example: Utility as $\sqrt{\text{Wealth}}$

A simple model: utility = $\sqrt{\text{dollars of net wealth}}$.

- Utility at \$100: $\sqrt{100} = 10$
- Utility at \$400: $\sqrt{400} = 20$
- Utility at \$900: $\sqrt{900} = 30$

Each extra \$300 gives you 10 more utility at first, but the jumps are worth less and less as you start wealthier.

Example: Utility as $\log_{10}(\text{Wealth})$

A slightly more realistic model (for wealth above \$1,000):

- Utility at \$10,000: $\log_{10}(10,000) = 4$
- Utility at \$100,000: $\log_{10}(100,000) = 5$
- Utility at \$1,000,000: $\log_{10}(1,000,000) = 6$

Notice that *multiplying* wealth by 10 only *adds* 1 to utility. The relationship between dollars and utility is deeply nonlinear.

A Worked Example

Suppose someone has a net wealth of \$100,000 (utility = 5).

They are offered a bet: 50% chance of winning \$900,000, 50% chance of losing \$90,000.

- Win: new wealth = \$1,000,000, utility = 6
- Lose: new wealth = \$10,000, utility = 4

$$\text{Exp}(U(\text{Bet})) = 0.5 \times 6 + 0.5 \times 4 = 5.0$$

Expected utility of the bet = 5 = status quo utility. They should be indifferent.

Reducing the Possible Gain

Now suppose the bet is: 50% chance of winning \$800,000, 50% chance of losing \$90,000.

- Win: new wealth = \$900,000, utility ≈ 5.95
- Lose: new wealth = \$10,000, utility = 4

$$\text{Exp}(U(\text{Bet})) = 0.5 \times 5.95 + 0.5 \times 4 = 4.98$$

The bet now has *negative* expected utility even though its *expected dollar return* is still very positive ($0.5 \times 800,000 - 0.5 \times 90,000 = \$355,000$).

Expected dollars and expected utility can point in opposite directions.

Insurance

The Insurance Puzzle

Every insurance contract is a bet, with you and the insurer on opposite sides.

The bet can't have a positive expected *dollar* return for both sides simultaneously.

So why does the insurance industry exist? Why is it rational for both sides to participate?

An Example

- A person has \$100,000 net wealth, including a car worth \$30,000.
- There's a 10% chance the car is destroyed in the next year.
- An insurer offers: pay \$3,200 now; if the car is destroyed, they refund its full value.

- 90% chance: wealth stays at \$100,000, utility = 5.0
- 10% chance: wealth falls to \$70,000, utility ≈ 4.845

$$\text{Exp}(U) = 0.9 \times 5.0 + 0.1 \times 4.845 = 4.984$$

Guaranteed wealth: $100,000 - 3,200 = \$96,800$.

$$U(\$96,800) = \log_{10}(96,800) \approx 4.986$$

$4.986 > 4.984$. The insured outcome has **higher expected utility**. Take the insurance.

The Insurer's Perspective

The insurer collects \$3,200 and pays out \$30,000 with probability 10%.

Expected dollar return = $3,200 - 0.1 \times 30,000 = \$200 > 0$.

As long as the insurer has a large, diversified pool of customers (so they effectively face the expected value, not the individual gamble), the insurance contract is profitable.

Why Both Sides Win

- The customer has a **concave** utility curve. They value certainty more than the raw dollar amounts suggest.
- The insurer, by pooling many independent risks, can afford to behave as if utility is approximately linear in money.
- This asymmetry makes mutually beneficial insurance possible.

Insurance isn't irrational, but this model also suggests people probably *over-insure* on small risks where the dollar amounts aren't life-changing.

The Allais Paradox

A Challenge to Expected Utility

Maurice Allais, a French economist, devised a pair of choices to challenge expected utility theory. The goal was to show that even committed expected utility theorists violate their own theory.

The choices are simple. The implications are not.

The Allais Paradox: First Choice

Which do you prefer?

- **1A:** \$1,000,000 for certain
- **1B:** 1% chance of \$0, 89% chance of \$1M, 10% chance of \$5M

Most people choose **1A**. The guaranteed million is too good to risk. The 10% shot at five million is not worth the 1% chance of walking away with nothing.

The Allais Paradox: Second Choice

Which do you prefer?

- **2A:** 89% chance of \$0, 11% chance of \$1,000,000
- **2B:** 90% chance of \$0, 10% chance of \$5,000,000

Most people choose **2B**. You'll probably walk away empty-handed either way, so the extra 1% risk seems worth it for the shot at five million.

Why This Is a Problem

Represent the options using 100 tickets:

	Ticket #1	Tickets #2–11	Tickets #12–100
1A	\$1M	\$1M	\$1M
1B	\$0	\$5M	\$1M
2A	\$1M	\$1M	\$0
2B	\$0	\$5M	\$0

Tickets #12–100 pay the same within each pair. The only comparison that matters is tickets #1 and #2–11. In those columns, 1A vs. 1B is *exactly the same* as 2A vs. 2B.

Let $x = U(\$5M)$, $y = U(\$1M)$, $z = U(\$0)$.

If $1A \succ 1B$: $y > 0.01z + 0.89y + 0.10x$

Simplifying: $0.11y - 0.10x - 0.01z > 0$

Now: $EU(2A) - EU(2B) = (0.11y + 0.89z) - (0.10x + 0.90z) = 0.11y - 0.10x - 0.01z$

The same expression. So $1A \succ 1B$ forces $EU(2A) > EU(2B)$. Choosing 1A over 1B and 2B over 2A is a contradiction, whatever your utilities are.

Maurice Allais (1911–2010) was a French economist. Writing at a time when many French intellectuals were skeptical of reducing decisions to a formula, he designed this example to prove Savage wrong.

Leonard Savage (1917–1971) was an American statistician and the leading advocate of expected utility theory.

When Savage first saw the problem, he made exactly the choices Allais predicted, violating his own theory. He then had to work out why those choices were wrong.

Allais thought the common choices ($1A \succ 1B$, $2B \succ 2A$) are *rational*, and that expected utility theory is therefore wrong.

The idea: there is something valuable about *certainty itself* that expected utility does not capture. Getting \$0 when you were one step away from a guaranteed million feels qualitatively different from getting \$0 in a situation that was already risky.

This “certainty effect” is a legitimate preference, Allais argued. Any theory that rules it out is missing something real.

The Sure Thing Principle

Savage's response is built on a principle:

Sure Thing Principle: *If you would choose X over Y knowing E is true, and you'd also choose X over Y knowing E is false, then you should choose X over Y even without knowing whether E is true.*

The reasoning: if E makes no difference to your comparison in either scenario, E is simply irrelevant. Your choice should not depend on it.

Apply the Sure Thing Principle with $E = \text{"ticket \#12-100 is drawn."}$

If E is true: 1A and 1B both pay \$1M, and 2A and 2B both pay \$0. No difference in either pair.

If E is false (ticket #1 or #2-11): the comparison is \$1M vs. (\$0 or \$5M), which is identical in both pairs.

So choosing 1A means you would not choose 2B in either scenario. By the Sure Thing Principle, you should choose 2A.

Prescriptive vs. Descriptive

There are two ways to use expected utility theory:

- **Descriptive:** predicts how people will actually choose (“most people will buy insurance because...”)
- **Prescriptive:** tells us how people *should* choose (“you ought to take the bet because...”)

Savage’s response is prescriptive. He grants that people are *tempted* to make the Allais choices. But being tempted does not make those choices rational.

The Allais paradox shows that actual choices can diverge from what expected utility recommends. That is not a problem for expected utility as a *normative* theory.

What to Make of This

- The Allais preferences are common in experiments. Both Allais and Savage agree on that.
- Whether they are rational is genuinely contested. Allais said yes; Savage said no.
- Modified theories (like Buchak's **risk-weighted expected utility**) can accommodate Allais preferences while remaining systematic.
- For this course, expected utility maximization is the standard framework. The debates around it are real and ongoing.

Practice Question

According to Savage's response to the Allais paradox, the rule of expected utility is:

- a. descriptive
- b. prescriptive
- c. both descriptive and prescriptive
- d. neither

Answer: (b). Savage treats expected utility as telling us how we *should* decide, not just describing how we do decide. The Allais preferences are common but irrational, on his view.

Practice Question

In the Allais Paradox, people tend to choose 1A over 1B, but 2B over 2A.

This suggests:

- a. for many people, the utility of \$5M is not five times that of \$1M
- b. people maximize expected utility, not expected monetary value
- c. people are more willing to accept a small risk when things are already risky
- d. the Sure-thing Principle is descriptive rather than prescriptive

Answer: (c). The context shifts between the two choices: a safe option is available in the first, but not in the second. That shift changes what people choose, even though expected utility says it shouldn't matter.

- We turn to **game theory**.
- What happens when the outcomes of your decisions depend not on nature, but on the choices of other *rational agents*?